

enhancing Heat transfer by controlling the near-wall turbulence - CHARGER

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This project proposes an investigative program to predict and enhance convective heat transfer through multi-scale flow control strategies. The first approach utilises wall oscillation to generate beneficial near-wall vortices that promote mixing in the inner region, where temperature gradient is steepest. The second strategy excites secondary flow instabilities such as Görtler vortices to induce large-scale structures that interact with the edge boundary layer, fostering thermal exchange between the near-wall flow and lower-temperature outer region. By synergistically combining actuation at these two critical scales, the aim is to maximise convective heat transfer while minimising drag penalties. This innovative dual-scale active flow control approach is expected to overcome limits of current heat exchanger designs. This fundamental research aligns with Néo Terra’s goals to increase renewable heat utilisation and aid the low-carbon transition of regional industries supporting decarbonisation efforts. The present proposal sets out a research programme that aims to explore heat transfer between a wall and a fluid through the triptych : “predict, model and control”.

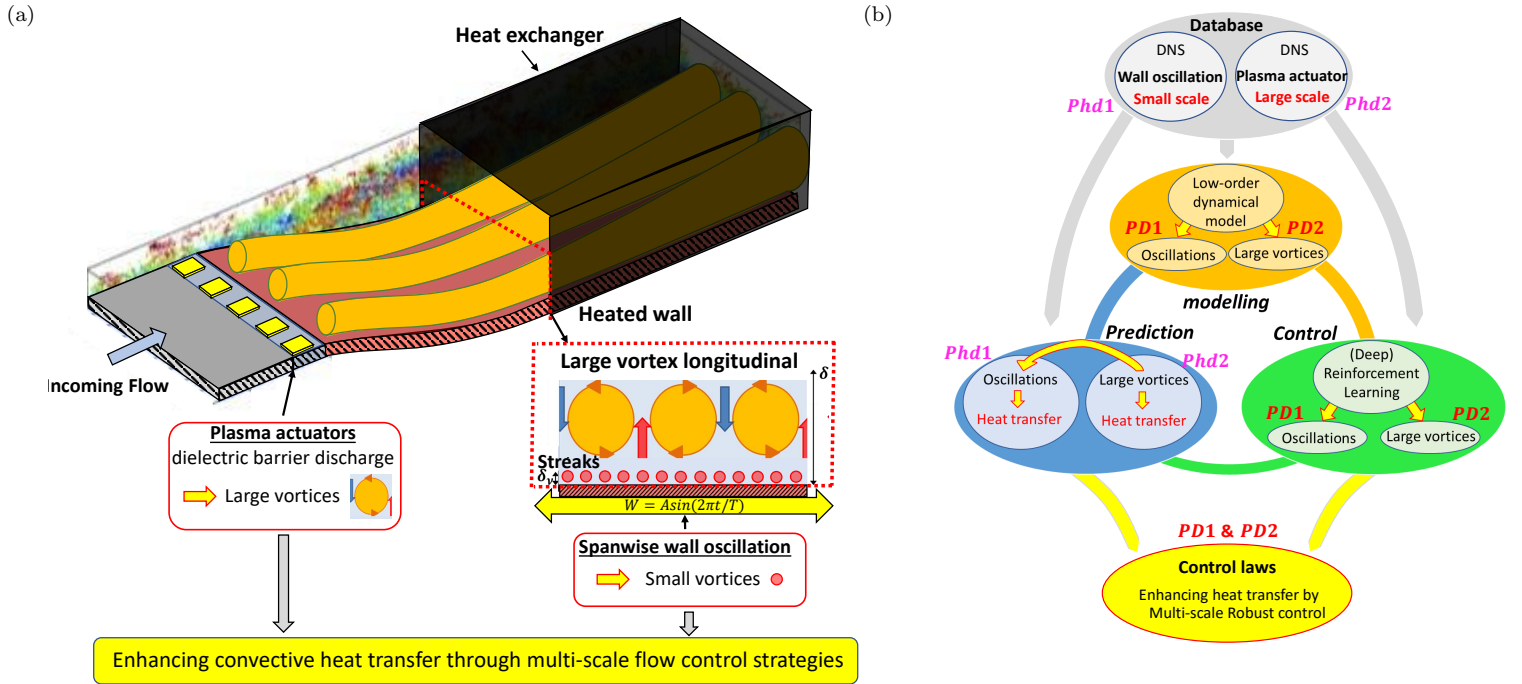


Figure 1: (a) Concept of high-efficiency heat exchanger enabled by active multi-scale control, leveraging flow instabilities. (b) Research program organisation

1 Context and objectives of the proposal

Enhancing heat exchanger performance is a critical engineering challenge across a wide spectrum of thermal applications. This challenge has become increasingly urgent in the context of rising global energy demands and the pressing need to transition towards sustainable and efficient energy systems. The optimisation of thermal energy conversion processes is particularly vital for renewable energy technologies.

Most thermal power generation systems rely on a two-step process: thermal energy capture and its subsequent conversion into electricity. Solar concentrators, for example, utilise concentrated solar radiation to heat a working fluid, which then drives a turbine to generate electricity. This dual-stage process is not limited to solar energy systems; it is also fundamental to other renewable energy sources, such as geothermal and biomass technologies. Improving the efficiency of these processes has the potential to revolutionise sustainable energy systems by significantly boosting energy output and reducing losses. Beyond these immediate technological advancements, optimising heat exchanger performance has far-reaching implications. It serves as a

strategic enabler for energy sovereignty by promoting localised renewable energy production, reducing dependency on external energy sources, and accelerating the decarbonisation of industrial processes. Such advancements directly contribute to combating climate change and fostering sustainable development.

This research programme aims to improve the efficiency of both steps, in line with regional targets to increase the use of renewable energy, by advancing knowledge in **modelling**, **control** and **prediction**:

- Recent advances in additive manufacturing are removing traditional design constraints, enabling fabrication of optimal heat exchanger geometries to enhance heat transfer. This “newfound freedom” necessitates fundamental research to understand the complex physics governing convective heat transfer in turbulent wall-bounded flows. Through detailed analysis of near-wall coherent structure modulation of the thermal boundary layer, advanced predictive **modelling** capabilities and physics-based design tools can be developed. Such innovations enable both passive flow optimisation through geometric design and active control through strategic manipulation of secondary instabilities.
- Heat transfer enhancement represents a persistent engineering challenge across diverse applications. Historical approaches predominantly employed passive control mechanisms, though active control strategies demonstrate superior performance potential despite implementation complexities. This research investigates the effectiveness of near-wall turbulence through **multi-scale active control strategies** to maximise heat transfer whilst maintaining acceptable drag characteristics. The proposed approach combines wall oscillation with strategic excitation of large-scale structures.
- Accurate **prediction** of temperature fluctuations is crucial for optimal system operation and component longevity, particularly in **gas turbines** for power generation and air transport. High-pressure turbine blades operate at temperatures exceeding material limits, necessitating sophisticated cooling systems. Enhanced thermal management could increase turbine inlet temperatures from 1530°C to 1830°C with the same cooling air volume, potentially delivering 30% greater engine power in aircraft gas turbines ([16]). However, prediction accuracy remains paramount - a 28°C error in metal temperature predictions can reduce blade lifespan by 50% ([16]).

The main objectives are : to gain insights on mechanism driving the heat transfer, to model and predict temperature evolution/fluctuations and finally propose a control strategy enhancing heat transfer but not at the expense of a large increase in drag. This is to be done by both **Experimental and computational investigations - aided by the application of machine learning**. The ML algorithms encompass, respectively, dimensionality reduction and low-dimensional dynamical modelling, and optimal control law design. Various techniques will be used to integrate physical knowledge into ML algorithms. Since they are physics informed, rapid convergence towards more robust solutions is achieved. The research does not specifically aim to facilitate the construction of improved heat-exchanger design. Rather, it entails a series of fundamentally-oriented studies on generic heated fluid subjected to idealised heating scenarios, the aim being to derive answers to basic questions on : **(i) the role plays by the streaky structures on the heat transfer, (ii) the effects of the outer-flow structures on the wall-temperature fluctuations, and finally (iii) the response of the flow to the proposed control methods in respect of heat transfer and drag**. In the nutshell, this project will use a unique combination of **numerical simulations, data-driven modelling and optimisation methods**, to fundamentally advance understanding of **heat transfer in wall-bounded turbulence**, and to propose and test **novel multi-scale active control strategies** to improve the **heat transfer efficiency and prediction**.

The proposed research is embedded within a broader collaborative framework involving IUSTI, Safran, and Pprime. No experiments will be conducted at Pprime due to the current limitations of its facilities, which are not equipped for this type of investigation. Consequently, a collaboration has been established with the IUSTI group, which possesses the necessary wind tunnel infrastructure and expertise in studying heated wall-bounded flows. Experimental work will be carried out at IUSTI, with experiments scheduled to begin in early 2025, supported by Safran funding. While the current project focuses exclusively on computational tasks, it will benefit significantly from the experimental measurements performed at IUSTI.

2 Position of the project as it relates to the state of the art

2.1 - Momentum and thermal boundary layers.

For any viscous fluid in motion relative to a solid, the velocity decreases to zero at the wall inducing a shear layer. Although emphatically chaotic, flow produced by this shear layer has some coherent structural components that are of major influence on the **momentum-mixing process** and consequently to the **drag** and **heat exchange**. The stronger the shear layer is, richer and more complex is the dynamics, which renders the study of near-wall Turbulence extremely fascinating and challenging. A conceptual representation of a boundary layer over a flat plate is conveyed in figure 2. When the gradient is relatively weak, only the viscous stress is in play. However, when the Reynolds number increases ($Re_L = U_\infty L/\nu$), the shear strain strengthens leading to the apparition of structures producing Reynolds stress, the flow departs from a laminar state to become turbulent. Turbulent flows dictate the performance characteristics of numerous industrial equipment and environmental applications. One important consequence of turbulence is high friction drag on surfaces, the increase relative to laminar conditions easily reaching factors of 10-100, depending on the Reynolds number of the flow. In many applications, the friction drag is extremely influential

to the operational effectiveness of the device or process. This applies especially to transport, involving either self-propelling bodies moving in a fluid or fluids being transported in ducts and pipes.

When a wall is at a temperature T_w , different from the temperature of the free stream, T_∞ , a thermal boundary layer is also present, with a thickness, δ_t , different from the momentum boundary layer thickness, δ . The turbulent thermal boundary layer has also an inner and outer regions. In the viscous sublayer, the turbulent mixing weakens drastically and conductive motions dominate (as illustrated in figure 2). Because turbulent mixing is ineffective in the viscous sublayer, the latter is known to contribute to a substantial fraction of the thermal resistance of a turbulent boundary layer. The buffer layer, just above the viscous sublayer, is the region where the Reynolds stress contributes the most, the heat transport is mainly driven by the turbulent mixing. Away from the wall, mixing motions gradually weaken until they ultimately vanish, defining the thickness of both thermal and momentum boundary layer. At first, it might seem that the Prandtl number (Pr defined in figure 2) is not associated to the heat transfer as δ_t and δ are similar, but even if Pr is associated to δ_t and δ it first describes the flow behaviour in the first part of the wall-bounded flow, where molecular viscosity and thermal conductivity still control the transport of heat and momentum. As the turbulent momentum mixing within the wall-bounded region enhances, the **thinner** the boundary layer becomes, offering **less resistance to heat flow but increasing the drag**. Thus, for heat exchangers, a balance need to be found between drag-induced losses and the heat transfer. To improve heat exchange, convection must be enhanced. The ratio of convective to conductive heat transfer across a boundary is defined by the Nusselt number (see definition of Nu_x in figure 2). By using the Reynolds-Colburn analogy ([8]), for a canonical turbulent boundary layer, the heat transfer can closely be related to shear stress on the wall by the semi-empirical equation 1. **One of the first objectives of the project CHARGER will be to extend equation 1 to actuated flow by wall oscillations, and latter investigate the effects of the Reynolds number and the presence of secondary flows.**

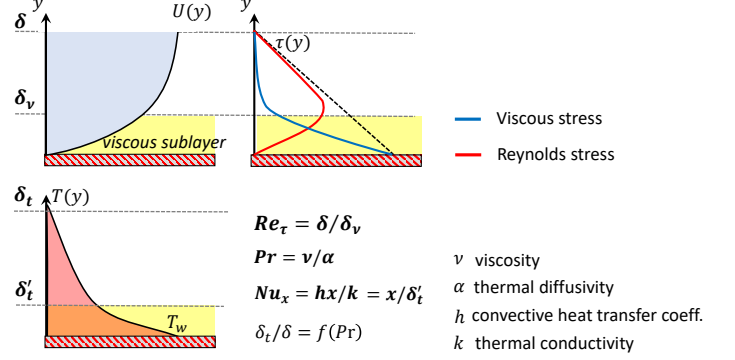


Figure 2: Conceptual representation of a momentum and thermal boundary layers.

$$Nu = \frac{Re Pr Cf/2}{1 + 12.7 (Pr^{2/3} - 1) \sqrt{Cf/2}} \quad (1)$$

2.2 Optimisation of heat transfer by manipulating inner-flow structures.

As previously mentioned turbulent boundary layer is populated with a wide spectrum of structures –“eddies”– that cover a range bounded by the Kolmogorov length, at one end, and multiples of the boundary-layer thickness, at the other. First, it is well established, both from simulations and experiments, that the highly-sheared viscosity-affected near-wall layer is characterised by a quasi-organised streaky structure – collectively referred to as “**streaks**” – formed by a spanwise, quasi-regular, elongated, high- and low-velocity regions. These are associated with quasi-streamwise vortices that cause a lift up of low-velocity fluid and a draw-down of high-speed fluid, thus important momentum mixing. Their development therefore leads to a strong drag increase, for decades the aerodynamic community has been investigating a wide range of passive and active control strategies to remove these structures. There are very few control options that are demonstrably effective at reducing friction drag **over large surfaces**, and it must be acknowledged that all involve formidable engineering challenges. Passive devices, such as riblets and super-hydrophobic surfaces, either offer modest drag-reduction margins and degrade rapidly, or are only realisable in very special circumstances - e.g. in liquids. Active methods, most of the time involve wall-based actuation. At low Reynolds number, all target the processes in the viscosity-affected near-wall layer, and all are designed to disrupt the near-wall streaky structure and associated regeneration mechanism that sustains the streaks and vortical motions. Leaving aside rather tentative attempts to use synthetic jets, active dimples/pimples and plasma actuators, mostly applied locally and at very low Reynolds number, one of tenable approach is to impart unsteady transverse shear onto the thin viscous sublayer by **oscillatory wall motion** ([7, 24]). The oscillatory flow patterns that are most effective involve streamwise waves of spanwise motion, where the spanwise velocity of the wall is $W = W_m \sin(2\pi x/\lambda_x - 2\pi t/T)$ ¹. With a careful choice of actuation parameters, gross drag-reduction level in excess of 45% has been demonstrated in direct numerical simulations of low Reynolds number channel flows [26, 25]. These studies show that the drag can be also substantially enhanced, depending on the choice of actuation parameters (see figure 3). It is established that for wall-scaled oscillation periods $T^+ \approx 100$ drag is substantially reduced. The exact mechanism by which such oscillations control the drag is complex and the subject of many studies (e.g. [27, 1]). Essentially, the oscillations set up an unsteady Stokes layer close the wall that weakens in the viscous sublayer and the quasi streamwise vortices, thus reducing turbulent momentum exchange close to the wall. An intensification of the near-wall turbulence can be expected to enhance the heat transfer, and a critical research question in the proposed programme is **whether it is possible to substantially increase the heat**

¹The control parameters are represented in red.

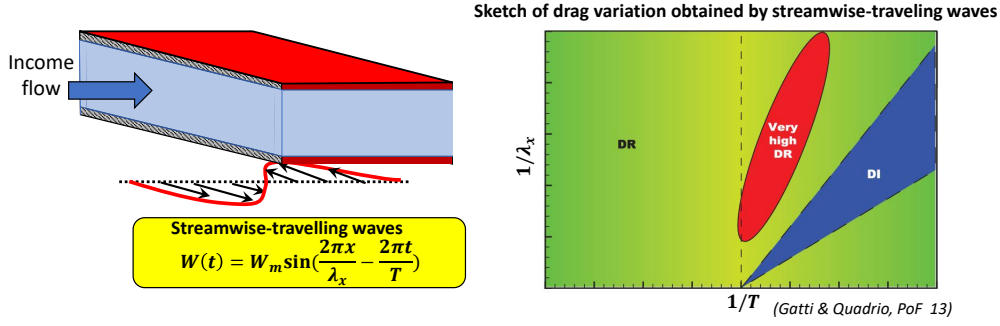


Figure 3: Left-hand side plot: schematic of the system for turbulent channel flow with travelling-wave actuation. Right-hand side plot: map of the drag variation in the plane spanned by λ_x - T , DR: Drag reduction and DI: Drag increase. ([13]).

transfer at only modest penalty of increased drag. The applicant and co-workers' preliminary computational results have demonstrated promising heat transfer enhancement using active flow control, without any optimisation. Specifically, heat transfer was increased by up to 15% with only a 7% drag penalty increase (paper under review). These early findings are highly encouraging, suggesting significant margins for further optimisation to achieve major thermal gains with minimal parasitic drag. The results highlight the potential of active control strategies to advance next-generation high-efficiency heat exchangers.

2.3 Optimisation of heat transfer by creating large streamwise vortices.

A broad range of methodologies has been examined to enhance heat transfer over a plate. These studies indicate that, irrespective of the type of turbulators employed, the devices demonstrating the highest efficiency are those that encourage counter-rotating secondary flows [17, 12]. The passive generation of longitudinal vortices has been proven to be notably effective at boosting heat transfer, as the vortices push outer flow towards the heated wall, where it gains thermal energy. The heated fluid is then transported back to the cooler outer region by the swirling motion. The vortices thus maximise temperature mixing and heat transport from the wall through the boundary layer. In the [9] study, the desired streamwise vortices were generated by combining vortex generators with relatively large parallel riblets perpendicular to the wall in the streamwise direction. With this set up, they were able to provide a solution for enhancing the heat transfer, however this design suffers from several limitations: (i) riblets result in significant parasite drag, inducing major pressure loss and thus reducing the efficiency of the turbine, (ii) riblets being fixed, the streamwise vortices generated cannot uniformly “pump-out” the heat over the entire surface. This limits the heat-transfer capacity and also cause temperature gradients in the material which might cause long term damage to the structure (to avoid structural failure, a balance must be made between conductivity and robustness of the material), and (iii) the large vortices can not effectively penetrate the viscous sublayer due to strong velocity gradient, thus they cannot by themselves efficiently extract the heat from the most heated part of the flow and transfer to the rest of the fluid.

Active flow manipulation strategies can be categorised into two distinct groups. The first encompasses techniques aiming to modify the primary flow, while the second group is based on either the introduction or the leveraging of secondary flows. When feasible, it is more desirable to utilise flow instabilities for flow control, as secondary flow movement naturally expands from a minor disturbance. To overcome the limitation associated to the use of riblet, or any other passive device, the applicants propose **to apply unsteady flow control strategies using periodic excitation that exploits natural flow instability phenomena**. Specifically, the project will utilise Görtler vortices which arise due to centrifugal forces on a curved surface, as illustrated in figure 1a. Periodic excitation by plasma actuators will harness and control these coherent structures to induce swirling motions for enhancing heat transfer. Active flow control is a rapidly growing multidisciplinary science and technology area aimed at altering a natural flow state towards a more desired condition. Active control for drag reduction has been extensively studied, but to the authors' knowledge, **active control for convective heat transfer enhancement in turbulent wall-bounded flows is still uncharted territory**. However, as momentum mixing drives both drag and heat transfer, knowledge gained from drag reduction strategies could be extended to heat transfer control. One innovative aspect of this research proposal is leveraging flow instabilities to generate large streamwise vortices with very weak perturbations. The benefits are: (i) they are easier to control, and (ii) more cost-effective. Unsteady flow control via periodic excitation harnessing natural flow instabilities has the potential to overcome efficiency barriers.

2.4 From physical understanding to Machine Learning approaches

The identification of the phenomena driving the flows dynamics remains one of the major challenges of physics and engineering today. This complexity has traditionally been addressed by statistical Turbulence models, but these are proven to have only limited generality, even in the case models are based on the most elaborate transport equations for the second moments of the turbulent fluctuations. The recent developments of **Machine Learning** methods for the **modelling of large-scale problems and their control** prove to be a promising way forward. Moreover, thanks to recent advances in metrol-

ogy, and the drastic reduction in computational costs, both the number and the size of database have undergone substantial growth in fluid mechanics. This abundance of data poses new challenges for the postprocessing that requires appropriate tools. Thankfully, many techniques have been developed by the **Machine Learning (ML) community** that are readily applicable to spatiotemporal fluid data, and are sustained by the recent advances in hardware (development of GPU/TPU platforms), algorithms and open-source libraries. By definition, ML is the field of study that gives computers the ability to learn without being explicitly programmed [28]. Machine Learning algorithms are categorised as *supervised*, *semi-supervised* and *unsupervised* learning. The task of supervised learning is to learn a function that maps input to output based on labelled examples. On the other hand, unsupervised learning consists in learning unknown patterns in a set of data without *a priori* information. Finally, semi-supervised learning operate on a large amount of unlabelled data, or with corrective information from the environment, for training a small amount of labelled data. In this project, Machine Learning algorithms will not aim to replace knowledge from physics (evolution laws for instance) with data, but rather to **use models, physical insights and data in synergy**. After years of in-depth analysis of boundary layer dynamics in compressible and incompressible regimes, and his recent studies using Machine Learning algorithms, the applicant believe that significant progress in understanding near wall flows and their control can only be achieved by making the best use of a solid background in turbulence and new approaches offered by Machine Learning.

2.5 Machine learning control: switch of paradigm

In a “*classical*” approach, the first step for determining an optimal control law is to fully understand the dynamics, then predict and finally optimise. However, for the vast majority of flow configuration, this “*sequential*” approach is untenable. Indeed, as flows are highly dimensional, strongly non-linear and multiscale, the first two phases required major effort, this is especially true for wall-bounded flows. Moreover, this model-driven approach has two major drawbacks: (i) it is difficult to interact in real time as the model is often complex, (ii) the dynamical model can not be easily generalisable. Machine Learning offers a shift of paradigm: the understanding, prediction and optimisation phases are an iterative, data-driven process, as shown in figure 4. At a given level of understanding, a prediction model is determined from the data, then a first optimisation strategy is evaluated, deepening our understanding and improving the prediction. Data-driven algorithms offer a most efficient approach for unravelling mechanisms driving flow dynamics and identify optimal control laws.

3 The interest of the project compared to the current state of the art, the needs of the economy, or potential applications.

3.1 Harnessing research for regional decarbonisation: Advancing low-carbon energy production technologies

This fundamental research on heat transfer and turbulence will enable major advances in heat exchanger technologies across sectors. The modelling and active control strategies can optimise thermal efficiency and adaptability, allowing adaption to fluctuating external conditions or tailoring to temperature requirements of downstream turbines. This directly benefits renewable systems such as solar concentrators and geothermal, reflecting regional priorities. The enhanced performance can also be integrated into heat exchangers for aviation engines and other thermal flow processes, supporting additional sustainability gains. The impact of thermal transfer is significant, encompassing renewable energy equipment, aerospace systems and industrial installations where thermal transfer is essential. Expected partnerships with regional industries will facilitate implementation of results into practical designs and sustainable technologies. For example, collaborations with NewHeat can be expected for incorporating insights into next-generation solar receivers. Joint efforts with Safran may improve aerothermal systems, furthering aviation decarbonisation. Furthermore, disseminating knowledge through collaborations with the R3 Tesna network and competitiveness centre will maximise the project’s impact, extending from industry to the public.

The project’s multidimensional strategy is in line with the regional priorities of Néo Terra in the renewable energy sector, as improving the fundamental knowledge of heat transfer will directly contribute to the development of new technologies that are crucial for low-carbon energy production, and collaboration with regional industry can accelerate the integration of these advances in key sectors, promoting a faster transition to a carbon-neutral economy in Nouvelle-Aquitaine and mix-energy goals.

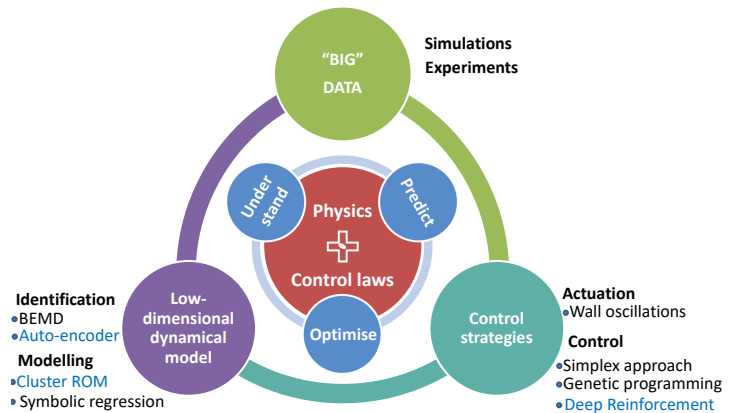


Figure 4: Machine learning control : switch of paradigm.

3.2 Structuring and emerging character of this research program

- the program aims to advance fundamental knowledge of heat transfer and turbulence physics, which can enable major improvements in renewable energy technologies like solar thermal. This aligns with regional priorities for sustainability and decarbonisation under Néo Terra.
- Partnerships with regional industries will facilitate implementation of research insights into practical designs and technologies. Collaborations can accelerate adoption of enhanced heat transfer methods in priority sectors.
- The project connects to Néo Terra’s goals for renewable energy and supporting the green transition of local companies. The expected performance gains address needs in solar, geothermal, aerospace, and other key regional industries.
- The project has an emergent quality, as success could position the region at the forefront of renewable thermal and heat transfer research. This can attract talent and future funding to the area .
- The proposal has a structuring aspect by bringing together public and private partners, and through plans for strong coordination and knowledge sharing.
- This project will strengthen and expand our collaboration with Safran. One of our key objectives is to establish a dedicated chair focused on heat exchangers, which will positively impact local employment and support the advancement and visibility of regional research initiatives. By focusing expertise on heat exchangers, the objective is to drive technological innovation and attract highly skilled talent, thereby stimulating economic and scientific development in the region.

3.3 Originality and the novelty

This project is **interdisciplinary**, with both fundamental science and practical engineering significance. It combines and advances the cutting edge of several advanced scientific concepts and capabilities; as such, it has the potential for substantial academic impact in broader contexts, regardless of ultimate practical use. First, the exploitation of advanced DNS for combined drag and heat-transfer optimisation, subject to a combination of active control strategies is novel and is likely to advance DNS into a new flow-control direction of interest to the flow-physics community. Second, the adaptation of Machine learning algorithms to optimise thermo-fluids systems is in its early infancy; the application of this emerging technology for modelling, predicting and optimising drag and heat transfer in the manner proposed has many unique aspects. The research outputs are expected to help model, predict and control heated wall-bounded flows.

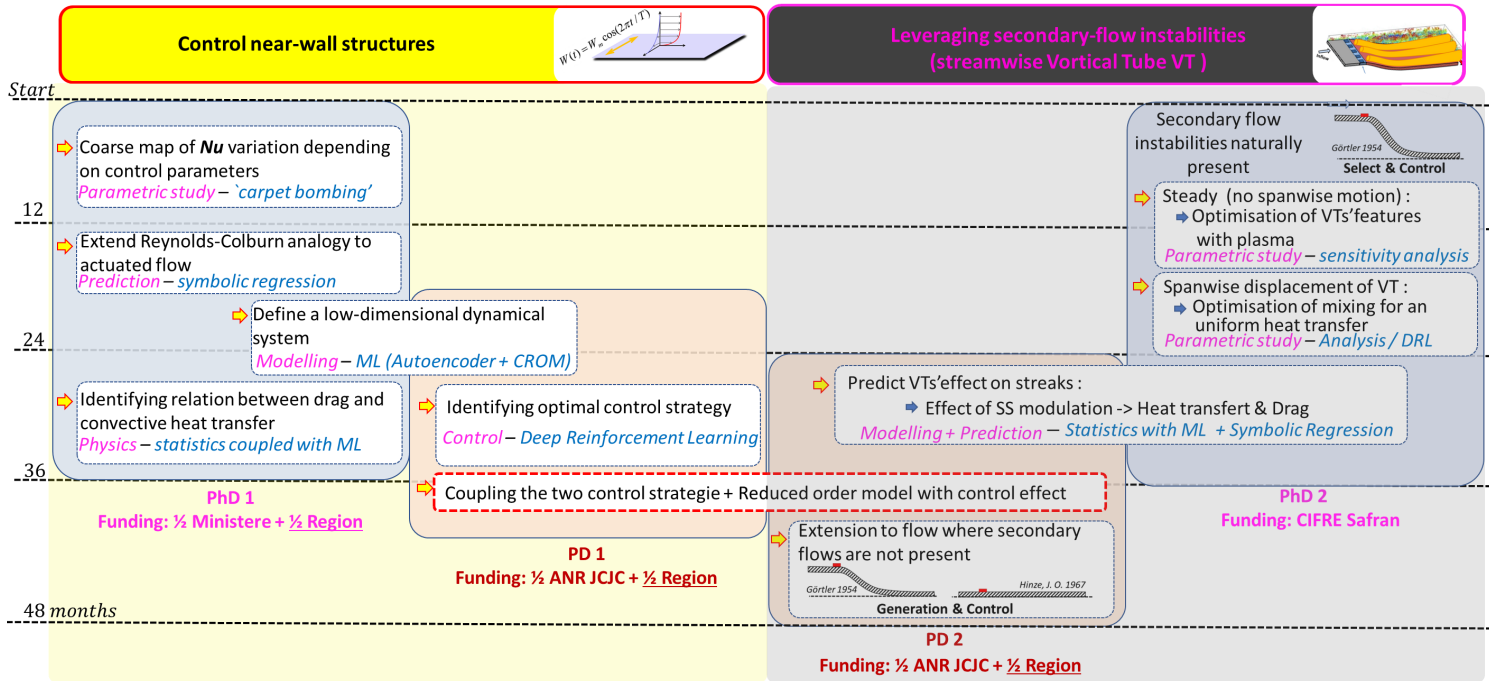


Figure 5: Gantt chart. It illustrates the timeline and division of research tasks between investigators, highlighting collaborative efforts on specific objectives. Each task links to a deliverable and associated publication. The primary goals of each task are indicated in magenta, while planned approaches are listed in blue.

4 Research program: Objectives, Methodology and risk management

4.1 Workpackages’ Objectives & Deliverables

The program is structured into five Work Packages (WPs), they combine **high-fidelity computational** investigations aided by the application by **data-driven** and **Machine Learning** (ML) approaches, with the aim of improving understanding of heat transfer propose and test realistic control strategies that can **maximise heat transfer** with a **minimum pressure-loss penalty**.

The data-driven modelling and ML approaches encompass, dimensionality reduction and low-dimensional dynamical modelling, and optimal control law design. The research programme’s structure and timeline are clearly outlined across two figures. Figure 1b illustrates how the different research tasks are connected, providing an overview of the project’s organisation. Meanwhile, Figure 5 presents a Gantt chart detailing the timeline and deliverables for each work package. The Gantt chart delineates individual boxes for each deliverable, with a minimum of one publication expected per deliverable. Helpfully, the text is colour-coded - black indicates the deliverable itself, magenta shows the type of study, and blue lists the methods to be developed. Beyond mapping out the timeline and expected deliverables, the Gantt chart also designates researcher responsibilities for each task, whilst highlighting dependencies between activities. Taken together, Figures 1b and 5 offer a comprehensive outline of the planned research programme, mapping its structure, timeline, deliverables, and team member roles in an intuitive visual format. The objectives and expected deliverables for each work package are outlined below :

- **WP1** investigates how small-scale turbulent structures influence heat transfer under spanwise wall oscillation for both drag increase and decrease cases, and optimise control laws. This **WP** will be divided into two subtasks:
 - **WP1a** focuses on analysis to relate small-scale turbulence statistics to convective heat transfer, and attempts to revise heat transfer correlations like the Nusselt number equation 1 by incorporating the effects of spanwise oscillation control. The main objective is to shed the light on : How do changes in near-wall coherent structures modify turbulent thermal transport? What modifications are needed to classical heat transfer models for actuated flow?
 - **WP1b** aims to determine optimal forcing parameters and develop effective control laws that enhance heat transfer while keeping pressure loss as low as possible. Challenges include identifying actuations that enhance heat transport without excessive drag penalties.

The goal of **WP1** is elucidate how small-scale structures and transport processes are altered by spanwise forcing, enabling optimised oscillation control law for heat-transfer enhancement.

- **WP2** leverages secondary instabilities (Görtler vortices) to enhance convective heat transfer.

- **WP2a** will define efficient vortical tube features, without spanwise motion, to maximise heat transfer and determine their footprint effects on near-wall turbulence. Key questions involve optimal vortical tube wavelength, amplitude, orientation, etc. for heat-transfer augmentation.
- **WP2b** will design control laws to sinusoidally move vortical tubes in the spanwise direction, homogenising temporal heat transfer. Challenges include finding effective flow perturbations to generate desired vortical tubes and preventing vortical tube decay during motion.
- **WP3** investigates how Görtler vortical tubes indirectly impact heat transfer and skin friction. This is achieved by analysing how the footprint of the large-scale vortical structures modifies the baseline small-scale turbulence features near the wall. The aim is to elucidate the specific mechanisms through which the presence of vortical tubes alters the statistics and coherent motions of the near-wall region. One of the main objective is to quantify the relative contributions of the direct versus indirect effects of the vortical tubes on turbulent heat and momentum transport. The study will then evaluate how the vortical tubes interfere with the efficiency of spanwise wall oscillations applied to control the small-scale structures, by comparing cases with and without actuation. This will provide insights into the complex interactions and modulation of the controlled near-wall turbulence by the large-scale vortical structures. The main objective of this WP is to investigate the effects of vortical-tube footprint on near-wall turbulence and determines the resulting indirect impacts on heat transfer, skin friction, and the efficiency of small-scale control.
- **WP4**. The main objective is to combine spanwise wall oscillation control of near-wall turbulence with Görtler vortical tube manipulation of outer layer coherent structures, in order to optimise convective heat transfer over a wide range of scales. This integrated approach leverages the capability of wall oscillations to efficiently modify the region of high temperature gradient, along with the effectiveness of large vortical tubes in enhancing mixing between the outer flow and near-wall region, where the temperature difference is the strongest. Building on understanding gained in the previous work packages. The main goals of this WP is to shed the light on: How can the two control methods be optimally combined to maximise heat-transfer enhancement across the full boundary layer? How do the two control strategies interact such that their effects are amplified rather than diminished? And finally, is it possible to identify and model the underlying mechanism that support the performance of the combined control?
- **WP5** aims to extend the combined control laws developed in **WP4** to flows that require perturbations to initiate secondary-flow instabilities. The goal are to identify effective methods for generating controlled secondary flows in real engineering systems and to overcome the additional complexity introduced by exterior perturbations. This will improve the robustness and expand the applicability of the integrated wall oscillation-plasma control framework. Major challenges to be addressed include determining optimal perturbation parameters, managing nonlinear interactions between vortical tubes and small-scale structures, and maintaining the optimal features of the secondary flows as they evolve spatially and temporally. Overcoming these issues will allow the control laws to be successfully implemented in real-world turbulent flows where secondary instabilities do not arise naturally.

The main objective of this research program is to advance fundamental understanding of how different scales in a turbulent boundary layer can enhance convective heat transfer, and propose **innovative multi-scale active flow control strategies as proof of concept demonstrations**. An important goal is to elucidate how near-wall coherent structures and large-scale coherent motions can be strategically manipulated to significantly improve thermal transport. This involves **developing predictive modelling capabilities and testing novel control protocols that target small-scales through spanwise wall oscillations and large-scales via secondary flow instabilities**. By combining complimentary actuation techniques, the objective is **enhancing heat transfer with minimum drag penalties**. Partnerships with regional industries are planned to facilitate technology transfer and integration of these modelling and control advances into practical renewable energy and aerospace systems. This will accelerate adoption in key sectors, fostering a faster transition to a carbon-neutral economy in Nouvelle-Aquitaine. This project **targets a TRL 4, but through collaborations the aim is to enable higher TRL demonstrations and eventual industrial implementation**. To summarise, this fundamental research promises to significantly advance knowledge of the physics of convective heat transfer, while delivering innovations that address regional priorities for decarbonisation.

4.2 Work Packages and Research Methodology: A Detailed Breakdown

WP1 - How small-scale structures manipulated by spanwise wall motions are efficient for driving the convective heat transport ? Efforts to optimise the heat transfer efficiency first require an in-depth fundamental understanding of how the small scales relate to heat transfer. This study will be undertaken through complementary numerical simulations by the PhD student and optimisation analysis by the Postdoctoral fellow :

⇒ **WP1a - Understand and predict the effect of the spanwise wall oscillations on the heat transfer**

- **PhD student 1** - Risk: **Low** for parametric study & **moderate** for predictive model of Nu .

Direct numerical simulations of channel flow will be performed using our in-house code to investigate drag reduction and heat transfer enhancement under spanwise wall oscillation. The simulations will be conducted at a low Reynolds number ($Re_\tau \approx 200$)

with the wall driven by sinusoidal motion, $W = W_m \sin(2\pi t/T)$, for various combinations of oscillation amplitude W_m and period T . The objective is to map the parameter space, similar to what is shown in figure 3, to elucidate the relation between drag and heat transfer under wall oscillations forcing. Preliminary results from a single parameter set [15, 14, see]

indicate the potential for substantial heat transfer augmentation, with an increase of 15% accompanied by only a 7% rise in pressure loss. However, it is important to note that this promising outcome is based on extremely limited data, as no parametric studies have been conducted and only this single combination of forcing parameters has been tested so far. Significant further exploration of the parameter space is needed to fully determine the tendency for maximising heat transfer while minimising excessive drag penalties. Nonetheless, these initial findings suggest strong potential for enhancing thermal transport through spanwise oscillation control.

Once an more extensive dataset mapping heat transfer over a wide range of oscillation amplitudes and frequencies is obtained, the objective is to revise the Reynolds-Colburn analogy given by equation 1 in order to develop heat transfer predictions accounting for the effects of spanwise oscillation control. To do so, we intend to use an approach based on **symbolic regression**. This latter is a ML algorithm able to learn to model input data with analytic expressions. The motivation of using symbolic regression is that the mathematical expression obtained is concise, interpretable and generalise better to out-of-distributing training data (as higher Reynolds number or higher wall temperature). For the research program we will use the regression method called **Gene Expression Programming (GEP)** is an *evolutionary algorithm* introduced by [11]. GEP is a purely data-driven method that was introduced in computer programming for solving complex optimisation problem. It is closely related to Genetic Programming (GP, see [20]) that is intensively used in the literature for determining closed-loop control laws [22]. The philosophy revolves around iteratively improving a population of candidate solutions by survival of the fittest. As with any evolutionary process, GEP relies on selection operators and random changes to the genome by means of mutation, and crossover. It is thus a non-deterministic algorithm. The general philosophy is to let the algorithm freely choose the best possible representation model by providing it only with a dictionary of unknowns and/or mathematical operations. The GEP outputs are then symbolic expressions corresponding to mathematical equations.

⇒ **WP1b - What oscillation parameters offer optimal enhancement of convective heat transfer through imposed wall motion ?**

- **Phd1 & Post Doc 1** - Progressive risk from **Low** to **High** depending of the complexity of the control law investigated.

A major advantage of using simulation over experiments for identifying efficient control approaches is the large spectrum of control strategies that can be investigated relatively quickly and at low cost. Indeed, the simulation route allows conditions to be investigated that are extremely difficult if not impossible to realise in a laboratory, thus exploring the extreme envelope of physically possible states. This promises a shorter route to gaining insight into the nature of the flow perturbation which must be implemented to achieve a realisable optimum. However, in order to be as efficient as possible, particularly in terms of computational costs, it is necessary to set up an appropriate framework. Here, we propose to use a **Reinforcement Learning (RL)** approach ([31]) to determine an optimal control strategy. RL is an autonomous, self-learning strategy that essentially acquires successive knowledge by interacting with its environment (here, the flow). In this approach an **agent** interacts with the **environment** through a series of **actions**. With every action, the agent modifies its environment and gets an immediate **reward**. The agent's goal is to determine those actions (the so-called *policy*) which maximise its long-term discounted rewards (minimisation or maximisation of energy, deviation from a target state known to be of physical interest, ...). Since RL is an evolutionary algorithm, a good compromise must be found between the *exploration* phase, during which the agent will gather more information leading it to better decisions in the future, and the *exploitation* phase where the agent will take the best decision based on the available information.

In this application, the control action is implemented through streamwise homogeneous wall oscillations. The key parameters that can be varied are the oscillation amplitude (W_m), frequency (T), and actuation waveform shape, which can be constructed from a summation of sinusoidal signals. Initially, studies will focus on identifying the optimal control law using a simple sinusoidal oscillation. Subsequently, research will determine the most effective waveform by optimising parameters including the number of harmonics (N), their amplitudes ($W_{m,i}$), frequencies (T_i), and relative phases (Φ_i). This two-step approach will systematically explore how oscillation characteristics influence turbulence modification and thermal transport, leading to enhanced convective heat transfer. Following [30], the policy will be represented by a Deep Neural Network (DNN) leading to **Deep Reinforcement Learning** (DRL=RL + DNN). Since the agent interacts with its environment in discrete time steps, **short-duration DNS simulations** for different actuation parameters will form a sequence of short runs, in each of which the flow will adjust rapidly (based on experience reported in [6]) to increments in the actuation. This will facilitate a rapid adjustment to the new actuation conditions, thus substantially decreasing the computational costs. To accelerate the convergence of the algorithm, we can imagine replacing the numerical simulations of the high-fidelity model (real environment) by an approximate model (reduced-order environment).

WP2 - How secondary-flow instabilities piloted by plasma actuators can improve large-scale convective transport ?

- **PhD student 2** - Progressive Risk to **Low** for parametric study to **moderate** for the use of deep reinforcement learning.

The objective of *WP* is to determine how secondary flow instabilities controlled by plasma actuators can enhance large-scale

convective heat transport. The simulations will be carried out using our in-house code.

The research will proceed in two stages :

- $\Rightarrow$ **WP2a** Optimisation of steady vortical tubes for maximum heat transfer. Systematic parametric studies will vary wavelength and amplitude to identify the optimal combination for heat transfer augmentation. Focusing initially on steady vortices will mitigate risks by limiting the parameter space exploration. Studying steady vortices will also aid analysis of small-scale turbulence modulation and the indirect heat transfer impact in **WP3**
- $\Rightarrow$ **WP2b** Introduction of controlled spanwise motion of the optimised VTs using plasma forcing. The aim is to induce uniform heat transfer distribution over the shortest timescale, while maintaining coherence of the structures. Parametric studies will find the actuation protocol that balances rapid transverse motion and preserving vortex length.

To mitigate the risk, a coarse parametric studies will be first be carried out. They will provide critical insights that can lead to more effective use of DRL for identifying optimal control strategies. Specifically, the coarse parametric studies can constrain and guide subsequent DRL exploration, limiting the parameter search space requiring heavy computational sampling. Identifying physics mechanisms, such as how specific flow structures enhance heat transfer, will allow incorporating this domain knowledge into the DRL reward/cost function design. Using physics-informed costs based on insights from the parametric studies will lead to more efficient convergence and avoid unproductive areas of the control space. In summary, prudent use of parametric studies to uncover dominant physics and constrain the control parameter space will enable better leveraging of DRL’s autonomous search capabilities. The combination of parametric mapping and DRL optimisation will lead to dramatic gains in identifying superior flow control strategies with feasible computational resources.

WP3 - How do large-scale VTs alter the small-scale near-wall turbulent structures and the efficiency of their control?

- PhD student 2 & Post Doc 2 - Progressive Risk to **Low** for statistic modelling to **moderate** local prediction using symbolic regression.

This work will focus on analysing the interaction between large-scale VTs and near-wall turbulence in wall-bounded shear flows, and leveraging this understanding to develop effective control strategies for heat transfer enhancement. Specifically, we will investigate how large eddies in the outer region of the boundary layer “footprint” on near-wall streaks and modulate the near-wall cycle through processes like amplitude modulation and increased intermittency.

The first step for investigating how the different families of turbulent structures contribute to heat transfer, is to separate the families from each others. To do so, we propose an data-driven methodology referred to as **“Auto-Encoder” (AE)**. An AE is an unsupervised machine-learning (ML) algorithm that is trained to reconstruct its inputs from a data set that is highly reduced in volume and detail relative to the full input. As illustrated in figure 6, an AE is a multi-layered neural network, composed of an encoder and a decoder, separated by a bottleneck. As a first step, the input data are compressed into a low-dimensional space by the encoder. The original input is then reconstituted by the decoder by using the compressed information embedded in the “latent” space. If some structure or order exists in the input data, in whatever sense (e.g. correlations between features), this structure is “learned” and consequently leveraged when forcing the input through the AE’s bottleneck. By reducing the latent space dimension, **AE has to exploit the natural data structure for identifying the most efficient representation**, in order to compress the data without any loss of important information. Dimensionality reduction can be seen as an **information-filtering process** as the size of the bottleneck constrains the amount of information allowed to traverse the full network. The low-dimensional representation of the flow dynamic within the latent space can also be leveraged for identifying a low-dimensional dynamical system. For the present WP, the AE is used in order to extract the effects of the VTs. Initially, the AE is trained to learn only the features associated with the outer-flow structures — i.e., the data used for training are streamwise-velocity-fluctuation fields at the wall-normal location where the large-scale outer flow structures are the most energetic. By reducing the amount of information passing through the AE’s bottleneck, the AE must learn the most significant features of the input so that the output image is as close as possible to the input image. In the present case, the level of closeness is defined by the absolute value of the reconstruction error, for steering the AE towards reconstructing the most energetic scales, which correspond to large-scale motions. Once the training (learning) is achieved, the AE will only use the

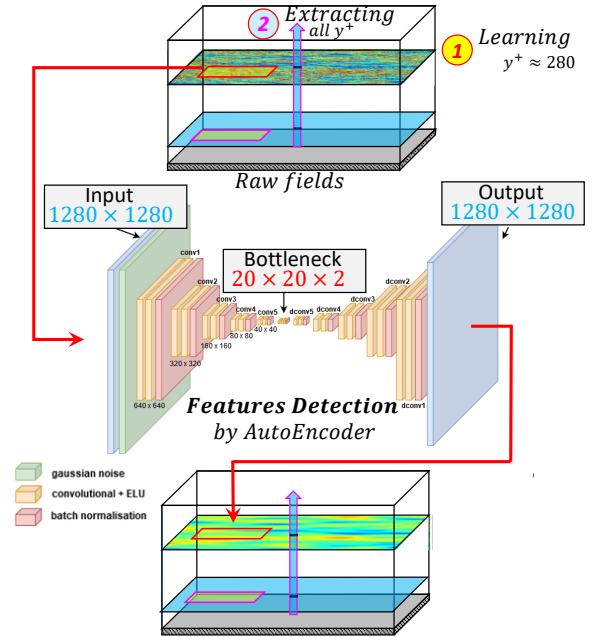


Figure 6: Schematic representation of the process of extracting the large-scale motions by using a multi-layer Auto-Encoder.

“library” of features learned in the outer-flow for identifying and extracting large-scale fluctuations generated by the outer-flow structures at every wall-normal location.

Once the large-scale structure are separated from the rest of the flow, one can determine how they alter the near-wall turbulence and the heat transfer. The first approach will be to performed a conditional statistical analysis based on multivariate joint pdfs [5, 4, see] to define a statistical predictive model and measure the influence of the control efficiency under the influence of VTs. The second approach is to use the symbolic regression algoritms to find a predictive model such the one developed for canonical boundary layer [23, 3, see].

By elucidating the interactions between VTs and inner scales, integrated multi-scale control laws that manipulate VTs and small structures in a synergistic way for optimum heat transfer enhancement can be developed. This dual scale approach, harnessing secondary flows and near-wall turbulence modulation, offers significant potential for thermal transport gains.

WP4 - How can various flow control strategies be integrated to create a multi-scale approach that enhances overall performance?

- **Post Doc 1 & Post Doc 2** - Risk :**Low** for coupling control strategies, progressive **moderate** to **high** for reduced order modelling

This work package provides an opportunity for close collaboration between the two postdoctoral researchers. The first goal is to synergistically integrate the two proposed flow control strategies. The second goal is to develop reduced order models that capture the essential physics of the control techniques with minimal computational cost. The merging of complementary actuation protocols and the creation of low-dimensional surrogates will enable feasibility studies for industrial implementation.

One common approach for obtaining a low-dimensional dynamical model is to determine a more relevant coordinate system in which the dynamics of the system can be represented in a simpler but more effective way. The modelling process can be divided into three steps: (1) reducing the problem dimensionality, (2) determining the elements of model’s skeleton within the reduced coordinate system, and (3) establishing the interactions between these elements. For the first two points, we intend to use Auto-Encoder technology [2] or a variante such as the Variational Auto-Encoder [10]. For the last part, the relationship between skeleton’s elements can be determined for example by applying symbolic regression methods [21], or the Cluster Reduced-Order Model (CROM) method [19]. Similar approach have already been used by the applicant for modelling a flow behind a cylinder [2], and we intend to use it in the present case, even if it a lot more challenging as the flow is more complex. The process is illustrated by Fig. 7. In the first step, we identify a low-dimensional projection subspace by using AE. As the flow features are coherent in space and time, a set of stacked snapshots is used as inputs of the Auto-Encoder. The encoder layer then converts the input data to encoded features embedded within an optimal subspace. In the second step, this low-dimensional representation is exploited to efficiently cluster the data set by similarity. By combining this low-dimensional representation of the data and a clustering algorithm (k-means for instance), a coarse grain description of the data is defined. Each cluster being associated to an ensemble of flow states, a “skeleton” of the low-dimensional space is obtained. In the third step, a dynamical model describing the transition probabilities from one cluster to another is derived by relying on a probabilistic Markov chain. Finally, the decoder layer is used to get a low-order dynamical representation of the data in the original space. This dynamical representation can greatly contribute to improve our understanding of the underlying dynamics.

WP5 - How can combined control strategies be extended to manage perturbation-induced secondary-flow instabilities in real engineering systems?

- **Post Doc 2** - Risk : progressive **moderate** to **high** for initiating and controlling secondary flow.

WP5 aims to build upon the combined control laws developed in WP4 by addressing flows that require external perturbations to initiate secondary-flow instabilities. These secondary flows are essential for enhancing mixing and heat transfer in various engineering applications. However, introducing such perturbations adds complexity, particularly in managing nonlinear interactions and maintaining flow stability. The primary objectives of WP5 are to develop strategies for generating controlled secondary flows within real engineering systems, improve the robustness of the integrated wall oscillation-plasma control framework against external perturbations, and extend its applicability to a wider range of turbulent flows where secondary instabilities do not naturally occur. To achieve these goals, the project will systematically determine the optimal parameters for perturbations that effectively initiate and sustain secondary flows. This involves utilising advanced computational models and machine learning techniques to manage and predict nonlinear interactions between vortical tubes and small-scale structures. In addition, feedback control mechanisms will be implemented to preserve the desired characteristics of secondary flows as they evolve spatially and

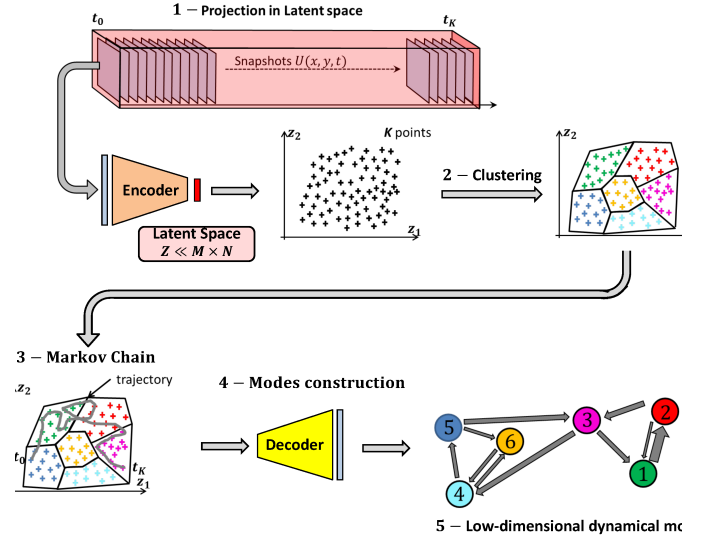


Figure 7: Reduced-order modelling framework that combines: dimensionality reduction by Auto-Encoders (step 1), clustering of the low-dimensional data (step 2), low-dimensional dynamical modelling of the encoded data (step 3), and decoded data (steps 4 and 5).

temporally. Experimental validation will play a crucial role in this process, allowing the refinement of theoretical models and control laws within controlled environments.

WP5 faces several major challenges, including finding the right balance of perturbation parameters to achieve the desired flow characteristics without introducing instability. Accurate modelling and controlling of complex nonlinear interactions within the turbulent flow is another significant hurdle. Furthermore, ensuring that secondary flows remain stable and effective over time and under varying operational conditions is essential for the successful implementation of the control laws. The expected outcomes of *WP5* include the development of a more robust and versatile control framework capable of handling perturbation-induced secondary flows in various engineering applications. In addition, the project aims to produce scalable solutions that can be adapted to different industrial settings, thereby improving the efficiency and performance of turbulent flow systems. Scientific advancements are also anticipated, particularly in the understanding of the connections between external perturbations and secondary-flow instabilities, which will contribute to the broader field of fluid mechanics and turbulence control.

5 Harnessing complementary expertise through effective task coordination in multidisciplinary teams

5.1 P.I. & Consortium specific expertise

Lionel Agostini is CNRS researcher at Pprime institute since October 2020. He defend his Phd on shockwave/boundary layer interaction at IUSTI (2011). His research work focus mainly on problems related to the presence of boundary layers, such as the drag reduction. The main purpose of his research is to unveil the physics and deepen the knowledge in order to provide more accurate flow-dynamic modelling and innovative control strategies. Over the writer's postdoctoral years at Imperial College London and Ohio State University, his studies have been oriented towards understanding and manipulating incompressible and compressible near-wall turbulence at moderate Reynolds number. In this period, the writer has gained extensive expertise in the area of wall turbulence. In this regard, he has developed and used a variety of statistics algorithms and data-driven methods for mining large datasets. His work in near-wall turbulence with and without actuation is well-known in the community. He has published widely in the field of aerodynamics – **23 publications (18 as first author) in first-ranked journals**. More recently, the applicant turns his attention on Machine Learning algorithms, and gained important expertise in this domain, and more specifically on Auto-Encoder (see [2, 4]). This knowledge will obviously be an extremely valuable asset to conduct the project. His previous work attests to his capacity for independent thinking. He has experience developing projects to address relevant hypotheses, planning its implementation and learning the necessary methodologies to gather and analyse data. He has incorporated a wide range of techniques into his skill set, making him competent to develop multidisciplinary integrative projects and to guide students in diverse, state-of-the-art questions. Exploiting this knowledge and insight combined with Pprime team expertise on: numerical simulations and data-driven methods, the author is in an excellent position to pursue **the objective of his research, to further our knowledge of near-wall turbulence, introduce more universal model and determine more efficient control devices**. This knowledge as long as the developed tools during the project, will be then leveraged for obtaining other funding and ultimately an ERC Grant. The PI will dedicate around 50% of his research time to the project. He will supervise the Phd students (36PM) and the Postdoctorate fellows (24PM). In addition of experts from Pprime, to successfully complete this project, the applicant can rely on several world-renowned experts.

In order to successfully complete this multi-disciplinary project, it is imperative to draw on the combined expertise of the different members of the team. The success of this research program will depend on the ability of the members to collaborate. The consortium will benefit of the expertise of **L.A.** and **Cédric Flageul (MCF)**, both expert in near-wall turbulence, **L.A.** is especially specialised in drag reduction and machine learning, while **C.F.** has important expertise in heat transfer and RANS simulations. The second Phd and Post-Doc will work with **Philippe Traore (Pr)** for the development and the use of ORACLE3D [29] which is an in-house code, design for fluid mechanic simulation with actuator plasma. For building low-order reduced order modelling and symbolic regression, the Phds and Postdocs will be able to relied on **Laurent Cordier (DR)** and **L.A.**'s expertise, s both have expertise in dimensional reduction and modelling. On top of that, **L.A.** has developed a large spectrum of tools, specifically designed for turbulent boundary layer. For the use of deep reinforcement learning, Phds and Postdocs will be guided by both **L.C.** and **P.T.**.

The experimental investigations conducted at IUSTI laboratory, under the supervision of **Sebastien Piponniau (MCF)** and **Pierre Dupont (CR)**, shall provide essential validation data for the numerical models and control methodologies developed within this project. The continuous hypoturbulent wind tunnel facility at IUSTI presents unparalleled capabilities for examining high-subsonic flows. The facility is equipped with advanced measurement techniques, including schlieren visualisation, steady and unsteady pressure probes, hot-wire anemometry, alongside sophisticated optical diagnostics such as Laser Doppler Anemometry (LDA), Particle Image Velocimetry (PIV) and Dual-PIV systems. These measurement techniques, combined with various optical access points, enable the acquisition of time-spatially resolved data under controlled experimental conditions [18]. The experimental data shall serve a dual purpose: firstly, to validate the numerical simulations and control strategies, and secondly, to guide the experimental team towards optimal control configurations by narrowing the parameter space for investigation. Furthermore, this synergistic approach between numerical and experimental studies shall inform the design specifications for flow control actuators, thereby enhancing the practical implementation of control strategies.

At Safran Aircraft Engines, the project will be supported by **Nicolas Fabbro** and **Thomas Bacoup**, a fluid systems experts

specializing in thermal management systems integration. **N.F** was especially involved in the certification of the CFM LEAP engine. **N.F** and **T.B** are now in charge of driving the roadmap related to computational methodologies and engineering tooling for the development of upcoming engines or demonstrators like the CFM RISE or the Airbus ZEROe program. Their expertise on integrated thermal management systems will be a useful asset to help carrying out this research project.

5.2 Effective Governance and Collaboration to Achieve Research Objectives

The advancement of the different tasks by the PhD students and post-docs will be presented every week during our weekly meetings, this weekly reporting and coordination will enable close monitoring of progress across all work packages, facilitating timely identification and resolution of any issues. Under the supervision of the PI, the students and researchers will collaborate within and between work packages to leverage complementary expertise and ensure synergy between activities. For instance, findings from *WP1* on small-scale flow physics will inform large-scale control approaches in *WP2*. The PI and co-workers' extensive experience leading projects and supervising students will ensure effective governance and completion of the ambitious research program. By incorporating insights from academic and industrial partners, the project aims to maintain focus on practical outcomes. The governance model balances autonomy for exploratory research with appropriate oversight and steering towards the overarching goals.

5.3 International Collaborations

The P.I can also expect insights from several world-renowned experts from various fields: numerical simulations [P. Vincent (ICL) and D. Gaitonde (OSU)], and Machine Learning [R. Vinuesa (KTH), F. Montomoli (ICL) and A. Mohan (LANL)]. Collaborations will be held in the form of regular visioconferences and occasional visits for the PhD student/Postdoc researcher, agreements have been found for waiving fees if necessary. The applicant also has funding resources for inviting some of them as visiting scholar to Pprime.

References

- [1] L. Agostini, E. Toubert, and M.A. Leschziner. "The turbulence vorticity as a window to the physics of friction-drag reduction by oscillatory wall motion". In: *International Journal of Heat and Fluid Flow* 51 (2015), pp. 3–15.
- [2] L. Agostini. "Exploration and prediction of fluid dynamical systems using auto-encoder technology". In: *Physics of Fluids* 32.6 (2020), p. 067103.
- [3] L. Agostini and M. Leschziner. "On the influence of outer large-scale structures on near-wall turbulence in channel flow". In: *Physics of Fluids* 26.7 (2014), p. 075107.
- [4] L. Agostini and Michael Leschziner. "Auto-encoder-assisted analysis of amplitude and wavelength modulation of near-wall turbulence by outer large-scale structures in channel flow at friction Reynolds number of 5200". In: *Physics of Fluids* 34.11 (2022).
- [5] L. Agostini and Michael Leschziner. "Statistical analysis of outer large-scale/inner-layer interactions in channel flow subjected to oscillatory drag-reducing wall motion using a multiple-variable joint-probability-density function methodology". In: *Journal of Fluid Mechanics* 923 (2021).
- [6] L. Agostini, E. Toubert, and M. A. Leschziner. "Spanwise oscillatory wall motion in channel flow: drag-reduction mechanisms inferred from DNS-predicted phase-wise property variations at $Re_\tau = 1000$ ". In: *Journal of Fluid Mechanics* 743 (2014), pp. 606–635.
- [7] F. Auteri et al. "Experimental assessment of drag reduction by traveling waves in a turbulent pipe flow". In: *Physics of Fluids* 22.11 (2010), p. 115103.
- [8] A. P. Colburn. "A method of correlating forced convection heat-transfer data and a comparison with fluid friction". In: *International Journal of Heat and Mass Transfer* 7.12 (1964), pp. 1359–1384.
- [9] A. Colleoni et al. "Optimization of winglet vortex generators combined with riblets for wall/fluid heat exchange enhancement". In: *Applied Thermal Engineering* 50.1 (2013), pp. 1092–1100.
- [10] H. Eivazi et al. "Towards extraction of orthogonal and parsimonious non-linear modes from turbulent flows". In: *Expert Systems with Applications* 202 (2022), p. 117038.
- [11] C. Ferreira. "Gene Expression Programming: a New Adaptive Algorithm for Solving Problems". In: *ArXiv Computer Science e-prints* (2001).
- [12] M. Fiebig. "Embedded vortices in internal flow: heat transfer and pressure loss enhancement". In: *International Journal of Heat and Fluid Flow* 16.5 (1995), pp. 376–388.
- [13] D. Gatti and Maurizio Quadrio. "Performance losses of drag-reducing spanwise forcing at moderate values of the Reynolds number". In: *Physics of Fluids* 25.12 (2013), p. 125109.

- [14] L. Guérin et al. “Preferential enhancement of convective heat transfer over drag via near-wall turbulence manipulation using spanwise wall oscillations”. In: *International Journal of Heat and Fluid Flow* 110 (2024), p. 109564. ISSN: 0142-727X.
- [15] L. Guérin et al. *Breaking the Reynolds Analogy: Decoupling Turbulent Heat and Momentum Transport via Spanwise Wall Oscillation in Wall-Bounded Flow*. 2023. arXiv: 2312.13002 [physics.flu-dyn].
- [16] J. C. Han, S. Dutta, and S. Ekkad. *Gas turbine heat transfer and cooling technology*. Second Edition. Taylor & Francis, 2012.
- [17] AM Jacobi and RK Shah. “Heat transfer surface enhancement through the use of longitudinal vortices: a review of recent progress”. In: *Experimental Thermal and Fluid Science* 11.3 (1995), pp. 295–309.
- [18] T. Jiang et al. “Velocity Spectrum Estimation in Shock-Wave/Turbulent Boundary-Layer Interaction”. In: *AIAA Journal* (2017/09/25 2017), pp. 1–13.
- [19] Eurika Kaiser et al. “Cluster-based reduced-order modelling of a mixing layer”. In: *Journal of Fluid Mechanics* 754 (2014), pp. 365–414.
- [20] John R Koza. “Genetic programming as a means for programming computers by natural selection”. In: *Statistics and computing* 4.2 (1994), pp. 87–112.
- [21] R. Li et al. “Drag reduction of a car model by linear genetic programming control”. In: *Experiments in Fluids* 58.8 (2017), p. 103.
- [22] R. Li et al. “Linear Genetic Programming Control for strongly nonlinear dynamics with frequency crosstalk”. In: *Arch. Mech.* 70 (6 2018), pp. 505–534.
- [23] I. Marusic, R. Mathis, and N. Hutchins. “Predictive model for wall-bounded turbulent flow”. In: *Science* 329.5988 (2010), pp. 193–196. (Visited on 12/09/2013).
- [24] Ivan Marusic et al. “An energy-efficient pathway to turbulent drag reduction”. In: *Nature communications* 12.1 (2021), pp. 1–8.
- [25] Maurizio Q., Pierre Ricco, and Claudio Viotti. “Streamwise-travelling waves of spanwise wall velocity for turbulent drag reduction”. In: *Journal of Fluid Mechanics* 627 (2009), pp. 161–178.
- [26] M. Quadrio and P. Ricco. “Critical assessment of turbulent drag reduction through spanwise wall oscillations”. In: *Journal of Fluid Mechanics* 521 (2004), pp. 251–271. (Visited on 01/22/2013).
- [27] P. Ricco et al. “Changes in turbulent dissipation in a channel flow with oscillating walls”. In: *Journal of Fluid Mechanics* 700.1 (2012), pp. 1–28. (Visited on 01/22/2013).
- [28] A. L. Samuel. “Some studies in machine learning using the game of checkers”. In: *IBM Journal of research and development* 3.3 (1959), pp. 210–229.
- [29] UK Seth et al. “Parametric study of a DBD plasma actuation based on the Suzen-Huang model”. In: *Journal of Electrostatics* 93 (2018), pp. 1–9.
- [30] D. Silver et al. “Mastering the game of Go with deep neural networks and tree search”. In: *Nature* 529.7587 (2016), p. 484.
- [31] R. Sutton and A. Barto. “Reinforcement Learning: An introduction”. In: (2011).